Iteration 1 (Abdul Kudus)

Part 3: Gradient Decent Manual Calculation setup and chain rule derivation

1. Using the model formula in matrix form

$$\hat{y} = m_1 x_1 + m_2 x_2 + b$$

$$\hat{y} = Xm + b$$

Initial given values

$$X = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix}, \quad y = \begin{bmatrix} 5 \\ 6 \end{bmatrix}, \quad m^{(0)} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \quad b^{(0)} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Row 1 of $X = (x_1, x_2) = (1, 3)$

Row 2 of $X = (4, 10)$ $\therefore n = 2$ data points

Bias $b = [b, b]^T$

Learning rate $\alpha = 0.01$

2. Using the Mean Squared Error (MSE) to find cost function

$$J(m, b) = \frac{1}{n} \sum_{i=1}^{n} (\hat{y}_i - y_i)^2 = \frac{1}{n} e^T e$$

$e = \hat{y} - y$ with $n = 2$

$$J = \frac{1}{2} \left[(\hat{y}_1 - y_1)^2 + (\hat{y}_2 - y_2)^2 \right]$$

3. Using the derivative chain rule

$$\frac{\partial J}{\partial \text{parameter}} = \frac{\partial J}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial \text{parameter}}$$

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(outer), using the power rule and chain rule,

$$\frac{\partial J}{\partial \hat{y}_i} = \frac{1}{n} \cdot 2(\hat{y}_i - y_i) \cdot 1 = \frac{2}{n} e_i$$

since $\frac{d}{du}(u^2) = 2u$ and $\frac{\partial}{\partial \hat{y}_i}(\hat{y}_i - y_i) = 1$

(Inner), using the constant-multiple rule on $\hat{y}_i = x_{i1} m_1 + x_{i2} m_2 + b$:

$$\frac{\partial \hat{y}_i}{\partial m_i} = x_{i1}$$

$$\frac{\partial \hat{y}}{\partial m_2} = x_{i2}, \quad \frac{\partial \hat{y}}{\partial b} = 1$$

multiply the inner and outer links

$$\frac{\partial J}{\partial m_1} = \sum_i \frac{2}{n} e_i x_{i1} = \frac{2}{2}(e_1 \cdot 1 + e_2 \cdot 4)$$

$$\frac{\partial J}{\partial m_2} = \sum_i \frac{2}{n} e_i x_{i2} = \frac{2}{2}(e_1 \cdot 3 + e_2 \cdot 10)$$

$$\frac{\partial J}{\partial b} = \sum_i \frac{2}{n} e_i \cdot 1 = \frac{2}{2}(e_1 + e_2)$$

Iteration matrix form

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e \qquad \frac{\partial J}{\partial b} = \frac{2}{n} 1^T e$$

$$X^T = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix}, \quad \frac{2}{n} = \frac{2}{2} = 1$$

Gradient decent update rule

$$m \leftarrow m - \alpha \frac{\partial J}{\partial m}, \quad b \leftarrow b - \alpha \frac{\partial J}{\partial b}$$

from the initial values: $m = [-1, 2]^T$, $b = 1$

Step 1: Using matrix product (row x column)

$$Xm = \begin{bmatrix} 1 & 3 \\ 4 & 10 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} (1)(-1) + (3)(2) \\ (4)(-1) + (10)(2) \end{bmatrix} = \begin{bmatrix} -1+6 \\ -4+20 \end{bmatrix}$$

$$= \begin{bmatrix} 5 \\ 16 \end{bmatrix}$$

Step 2: Using the model formula for predictions

$$\hat{y} = Xm + b \qquad \hat{y} = \begin{bmatrix} 5+1 \\ 16+1 \end{bmatrix} = \begin{bmatrix} 6 \\ 17 \end{bmatrix}$$

Step 3: Error $= e = \hat{y} - y$

$$e = \begin{bmatrix} 6-5 \\ 17-6 \end{bmatrix} = \begin{bmatrix} 1 \\ 11 \end{bmatrix}$$

Step 4: Cost with MSE $= J = \frac{1}{n} \sum e_i^2$

$$J = \frac{1}{2}(1^2 + 11^2) = \frac{1}{2}(1 + 121) = \frac{122}{2} = 61$$

Step 5: Gradient with respect to m using Chainrule

$$\frac{\partial J}{\partial m} = \frac{2}{n} X^T e$$ matrix multiplication;

$$X^T e = \begin{bmatrix} 1 & 4 \\ 3 & 10 \end{bmatrix} \begin{bmatrix} 1 \\ 11 \end{bmatrix} = \begin{bmatrix} (1)(1) + (4)(11) \\ (3)(1) + (10)(11) \end{bmatrix} = \begin{bmatrix} 1+44 \\ 3+110 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

$$\frac{\partial J}{\partial m} = 1 \times \begin{bmatrix} 45 \\ 113 \end{bmatrix} = \begin{bmatrix} 45 \\ 113 \end{bmatrix}$$

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Step 6: Gradient with respect to b

$$\frac{\partial J}{\partial b} = \frac{2}{n}(e_1 + e_2)$$

$$= \frac{\partial J}{\partial b} = 1 \times (1 + 11) = 12$$

Step 7: Using the gradient update rule for updating

new = old − α × gradient

$m_1 = -1 - 0.01(45) = -1 - 0.45 = 1.45$

$m_2 = 2 - 0.01(113) = 2 - 1.13 = 0.87$

$b = 1 - 0.01(12) = 1 - 0.12 = 0.88$

$b = [b, b]^T$

$b = [0.88, 0.88]^T$

Result for iteration 1: $m = [-1.45, 0.87]^T$,

$b = 0.88$

Iteration 2 (Bintia)

Date / /

Starting Values (Iteration 1 Result)

$$m = [-1.45, 0.87]^T, b = 0.88,$$

Step 1 - Matrix product xm

$$xm = \begin{bmatrix} (1)(-1.45) + (3)(0.87) \\ (4)(-1.45) + (10)(0.87) \end{bmatrix} = \begin{bmatrix} -1.47 - 26 \\ -5.80 - 87 \end{bmatrix}$$

$$= \begin{bmatrix} 1.16 \\ 2.90 \end{bmatrix}$$

Step 2 - predictions

$$\hat{y} = \begin{bmatrix} 1.16 + 0.88 \\ 2.90 + 0.88 \end{bmatrix} = \begin{bmatrix} 2.04 \\ 3.78 \end{bmatrix}$$

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Step 3 : ERROR $e = \hat{y} - y$

$$e = \begin{bmatrix} 2.04 - 5 \\ 3.78 - 6 \end{bmatrix} = \begin{bmatrix} -2.96 \\ -2.22 \end{bmatrix}$$

step 4 : Cost - $J = \frac{1}{n} \sum_i e_i^2$

$$J = \frac{1}{2}\left((-2.96)^2 + (-2.22)^2\right)$$

$$= \frac{1}{2}(8.7616 + 4.9284)$$

$$= \frac{1}{2}(13.69)$$

$$= 6.845$$

Step 5 - Gradient with m

$$X^T e = \begin{bmatrix} (1)(-2.96) + (4)(-2.22) \\ (3)(-2.96) + (10)(-2.22) \end{bmatrix}$$

$$= \begin{bmatrix} -2.96 - 8.88 \\ -8.88 - 22.20 \end{bmatrix}$$

$$= \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

Date / /

$$\frac{\partial J}{\partial m} = 1 \times \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix} = \begin{bmatrix} -11.84 \\ -31.08 \end{bmatrix}$$

Step 6: Gradient to b

$$\frac{\partial J}{\partial b} = 1 \times (-2.96 - 2.22) = -5.18$$

Step 7: Update

$$m_1 = -1.45 - 0.01(-11.84) = -1.3316$$

$$m_2 = 0.87 - 0.01(-31.08) = 1.1808$$

$$b = 0.88 - 0.01(-5.18) = 0.88 + 0.0518 = 0.9318$$

Results = $m = [-1.3316, 1.1808]^T$,

$b = 0.9318$

ITERATION 3. - BIOR MAJOK

Starting values (result of Iteration 2 by Binthia:

$$m = [-1.3316, 1.1808]^T, \quad b = 0.9318$$

Step 1. Matrix product Xm = using matrix multiplication (row x colum

$$Xm = \begin{bmatrix} (1)(-1.3316) + (3)(1.1808) \\ (4)(*4(-1.3316) + (10)(1.1808 \end{bmatrix} = \begin{bmatrix} -1.3316 + 3.5424 \\ -5.3264 + 11.808 \end{bmatrix}$$

Step 2. Prediction - Using the model formula $y = Xm + b$

$$\hat{y} = \begin{bmatrix} 2.2108 + 0.9318 \\ 6.4816 + 0.9318 \end{bmatrix} = \begin{bmatrix} 3.1426 \\ 7.4134 \end{bmatrix}$$

Step 3 - Error - Using $e = \hat{y} - y$

$$e = \begin{bmatrix} 3.1426 - 5 \\ 7.4134 - 6 \end{bmatrix} = \begin{bmatrix} -1.8574 \\ 1.4134 \end{bmatrix}$$

Step 4 - Cost - Using the MSE formula $J = \frac{1}{n}\sum e_i^2$

$$J = \tfrac{1}{2}((-1.8574)^2 + (1.4134)^2 = \tfrac{1}{2}(3.44993476 + 1.997699956) = \tfrac{1}{2}(5.447634432) = 2.723817216.$$

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Step 5 = Gradient with respect to m = Using the chain-rule

result. $\frac{\partial J}{\partial m} = \frac{2}{n} X^T e$

$$X^T e \begin{bmatrix} (1)(-1.8574) + (4)(1.4134) \\ (1)(-1.8574) + (10)(1.4134) \end{bmatrix} = \begin{bmatrix} -1.8574 + 5.636 \\ -5.5722 + 14.134 \end{bmatrix} =$$

$$\begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix}, \quad \frac{\partial J}{\partial m} = 1 \times \begin{bmatrix} 3.7962 \\ 8.5618 \end{bmatrix} = \begin{bmatrix} 3.7962 \\ 8.8618 \end{bmatrix}$$

Step 6. Gradient with respect b - Using the chain rule

$$\frac{\partial J}{\partial b} = \frac{2}{n} (e_1 + e_2)$$

$$\frac{\partial J}{\partial b} = 1 \times (-1.8574 + 1.4134) = 0.444$$

Step 7 - Update - Using the gradient descent rule

$M_1 = 1.3316 - 0.01(3.7962) = -1.3316 - 0.037962 = -1.36956$

$M_2 = 1.1808 - 0.01(8.5618) = 1.1808 - 0.085618 = 1.095182$

$b = 0.9318 - 0.01(-0.444) = 0.00444 = 0.93624$

Result of Iteration 3 (final, passed to kvdvs for the summary):

$m = [-1.369562, 1.095182]^T$, $b = 0.093624$

Check of the final values - Using step 1-4 again with final m and b

$$\begin{bmatrix} -1.369562 + 3.285546 + 0.93624 \\ -5.478248 + 10.95182 + 0.93624 \end{bmatrix} = \begin{bmatrix} 2.85224 \\ 6.409812 \end{bmatrix}$$

$$e = \begin{bmatrix} -2.147776 \\ 0.409812 \end{bmatrix}$$

$$J = \frac{1}{2}(4.612941746176 + 0.167945875344) =$$

$$\frac{1}{2}(4.780887621520) = 2.390443810$$

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Gradient Decent Summary and Trend

Learning rate $\alpha = 0.01$

Iteration (by)	m at start	b at start	g	e	J	∂J/∂m	∂J/∂b	m result	b result
1 (Kudus)	[-1, 2]	1	[6, 17]	[1, 11]	61	[45, 113]	12	[-1.45, 0.87]	0.88
2 (Binthia)	[-1.45, 0.87]	0.88	[2.04, 37]	[-2.96, -2.22]	6.845	[-11.84, -31.08]	-5.18	[-1.3316, 1.1808]	0.9318
3 (Bior)	[-1.3316, 1.1808]	0.9318	[3.1426, 7.4134]	[-1.8574, 1.4134]	2.7238	[3.7962, 8.5618]	-0.444	[-1.369562, 1.095182]	0.93624

Cost after Iteration 3: J = 2.39044381

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Part 3. Question 7

Describe the trend in m and b. Are they moving in a direction that reduces the error?

summary

Answer: From the table, error after iteration 3

$\bar{J} = 2.3904$

Yes, the error $\bar{J}$ falls after every iteration (61 → 2.39)

Trend

1. The error goes down every time:
   61 → 6.845 → 2.7238 → 2.3904 (96% smaller)
2. Each value moves opposite to its gradient (downhill) against the sign of its gradient:
   Iteration 1: positive gradients → value decreases
   for gradients +45, +113, +12 → m_1, m_2, b all went down.
   Iteration 2: Negative gradients → value increases
   for gradients -11.84, -31.08, -5.18 → all went up
3. The steps get smaller as the error gets smaller: change in Δm_2: 1.13 → 0.3108 → 0.0856
   and the first step is the largest because the first error was the largest ($\hat{y}_2$=17 vs 6)

4. m_2 zig-zags but settles (2 → 0.87 → 1.1808 → 1.0952) and the error of point 2 changes sign (+11 → 2.22 → 1.41) : $x_2 = 10$ makes the cost very steep along m_2, so $\alpha = 0.01$ slightly overshoots, but each overshoot is smaller than the last one before

In conclusion, m and b are moving in the direction that reduces the error. The error fell 96% in 3 iterations and because each step is smaller than the last, the values are settling down. If we kept iterating, the error would keep falling towards 0: with only 2 data points and 3 parameters (m_1, m_2, b), a line that passes through both points exists, so the smallest possible error is 0